# LOGICAL EQUIVALENCE

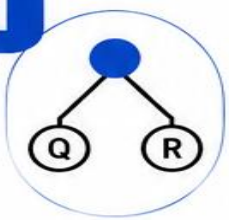
## LAWS OF PROPOSITIONAL LOGIC

### WHAT IS LOGICAL EQUIVALENCE?

Two logical expressions are equivalent if they have the **same truth value** in every possible situation.

| P | Q | P ∧ Q |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | F |
| F | F | F |

### 1 DISTRIBUTIVE LAW




| FORMULA | EXPLANATION | EXAMPLE |
|---|---|---|
| $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$<br>$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$ | • AND distributes over OR, and OR distributes over AND. | $A \wedge (B \vee C)$ ↓ $(A \wedge B) \vee (A \wedge C)$ |

### 2 DE MORGAN'S LAWS

$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$

$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

- Negation flips AND to OR, and OR to AND.

NOT(A ∧ B) ↓ NOT A ∨ NOT B

### 3 ABSORPTION LAW

$P \vee (P \wedge Q) \equiv P$

$P \wedge (P \vee Q) \equiv P$

- Extra parts that are already included do not change the result.

A ∨ (A ∧ B) ↓ A

### 4 COMPLEMENT LAW

$P \vee \neg P \equiv \text{True}$

$P \wedge \neg P \equiv \text{False}$

- A statement with its negation is always a constant.

A ∨ ¬A ↓ True

### 5 IMPLICATION

$P \rightarrow Q \equiv \neg P \vee Q$

- "If P then Q" can be rewritten using OR.

If A → B ↓ NOT A ∨ B

### 6 BICONDITIONAL

$P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$

- Both statements must have the same truth value.

A ↔ B ↓ (A → B) ∧ (B → A)

## WHY IT MATTERS IN COMPUTER SCIENCE

**1. LOGIC CIRCUITS**
Simplifies and optimizes digital circuit design.

**2. PROGRAMMING**
Helps simplify conditions and improve code efficiency.

**3. DATABASES**
Improves query optimization and data retrieval.

**4. SECURITY**
Used in encryption, access control and security protocols.

**5. ARTIFICIAL INTELLIGENCE**
Foundation for reasoning, inference and decision-making systems.

**IN SHORT:** These laws help us transform, **simplify** and verify logical expressions without changing their meaning.